

Transfer Capabilities of BlendGAN by Leveraging InterFaceGAN

Ali Osman Berk Şapcı · Cem Kaya · Furkan Reha Tutaş

Sabancı University, 2022 · typeset September 2026

Abstract

Generative adversarial networks have improved rapidly in recent years. This report focuses on two of them. InterFaceGAN performs semantic face editing by interpreting the latent semantics a GAN has learned; BlendGAN generates arbitrarily stylised faces through a flexible blending strategy trained on a large artistic-face dataset. We combine the two: a face is edited in the GAN’s latent space with InterFaceGAN and the edited face is then stylised with BlendGAN from a single reference image. We compare three ways of encoding a real photograph into the editable latent space, and analyse how well BlendGAN preserves the edited attributes (age, eyeglasses, gender, smile, head pose) under stylisation. Animated versions of the attribute sweeps are in the repository README: github.com/specialone0007/transferCapabilitiesOfBlendganByLeveragingInterfacegan.

1 Background

1.1 Generative adversarial networks

GANs generate high-quality photo-realistic images by adversarial training between a generator and a discriminator; that competition is the key to their success and the reason for the attention they have received.

1.2 BlendGAN

BlendGAN (Liu et al., 2021) generates stylised face images of high quality:

1. A style encoder learns a style code from a reference style image; the codes are normally distributed.
2. Two MLPs map the face code and the style code into separate spaces.
3. A weighted blending module (WBM) combines the two spaces and feeds the generator, which produces both natural and stylised faces.
4. Three discriminators judge realism: a face discriminator (real vs. generated natural face), a style discriminator (real vs. generated stylised face) and a style-latent discriminator (stylised face vs. its style code).

1.3 InterFaceGAN

InterFaceGAN (Shen et al., 2020) turns an unconditionally trained face generator into a controllable one:

1. Take a pretrained GAN and an attribute scorer.

2. Sample latent vectors, generate faces, score the attributes of interest.
3. Fit a linear SVM in latent space for each attribute; the normal of its separating hyperplane is the editing direction.
4. Edit a face by moving its latent code along that normal.

Because the best pretrained boundaries are for StyleGAN, there are several latent spaces ($\mathcal{Z}$, $\mathcal{W}$, $\mathcal{W}^+$) in which an image can be encoded.

2 Problem

Semantically edit a given face using the latent space of a GAN, then stylise the edited face for a given art style, and analyse the attribute-transfer capabilities of BlendGAN when it is driven by InterFaceGAN edits.

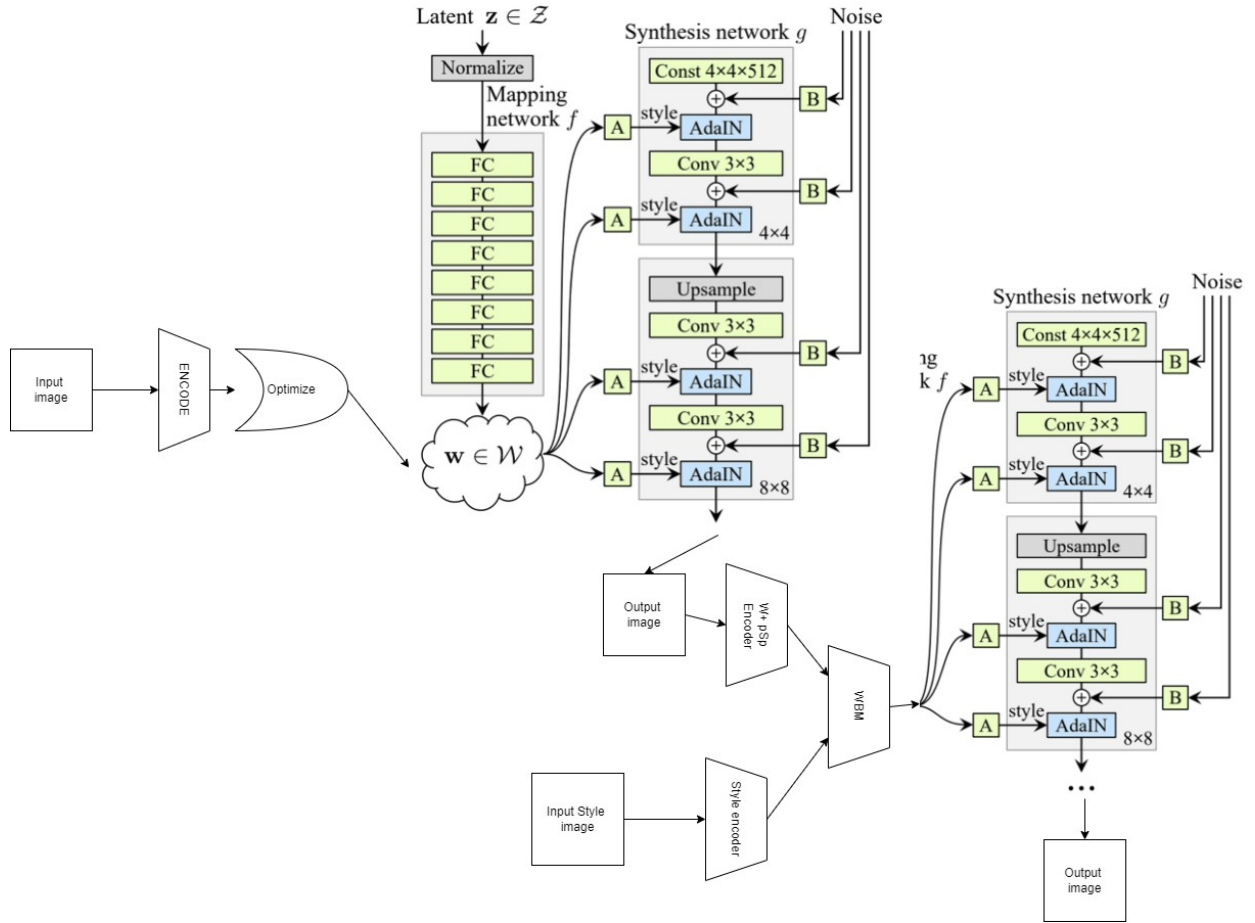


Figure 1: Proposed pipeline. A photo is encoded (encoder, optimisation or both) into StyleGAN’s latent space, edited with InterFaceGAN boundaries, rendered, and finally stylised by BlendGAN with a reference style image.

InterFaceGAN edits gender, age, expression, eyeglasses and head pose, and can also repair artefacts in generated images. BlendGAN stylises the result from a reference image and lets us draw on the AAHQ artistic-face dataset released with it.

3 Encoding a photograph into the editable latent space

InterFaceGAN edits *sampled* faces well, but editing a real photograph needs GAN inversion first. Following the taxonomy of Xia et al. (*GAN Inversion: A Survey*) we tried three routes.

3.1 Approach 1: a trained encoder

A ResNet-38 was trained to regress the $\mathcal{W}$ code (1×512) from 50,000 image/latent pairs. Its reconstructions are semantically similar but neither photo-realistic nor the same person. pSp encoders (pixel2style2pixel, which use the generator as the decoder of an auto-encoder) were trained on the commercially usable portion of FFHQ (52,000 images) but did not converge within our budget: 3.2 hours per epoch at batch size 4 on an 8 GB GPU.

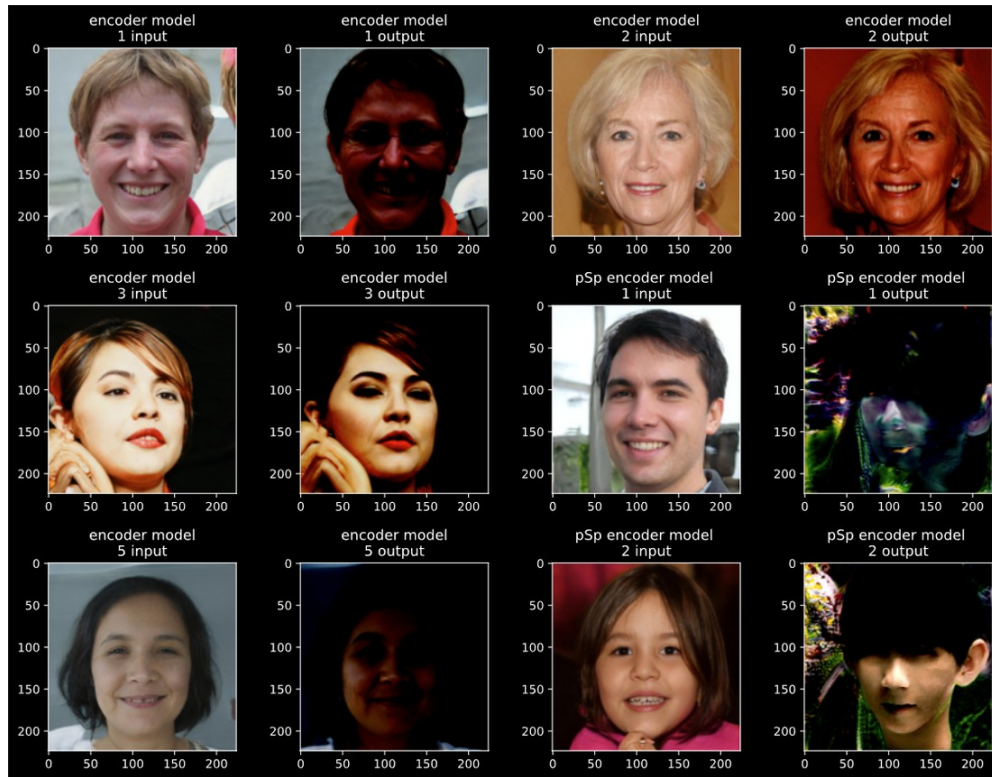


Figure 2: Trained-encoder reconstructions (input / output pairs) for the ResNet and pSp encoders.

3.2 Approach 2: latent optimisation

Start from a random $\mathcal{W}^+$ code (18×512), which the survey shows can give high-fidelity reconstructions, and optimise it by back-propagating a reconstruction loss through the generator. Without a close starting point the results are not photo-realistic, and each image takes about five minutes.

3.3 Approach 3: hybrid

Use the encoder’s prediction as the starting point and then optimise. This gave photo-realistic, semantically close reconstructions. Identity is not fully preserved, but the resulting codes are **stable under semantic editing**, which is not a given for every inversion method.

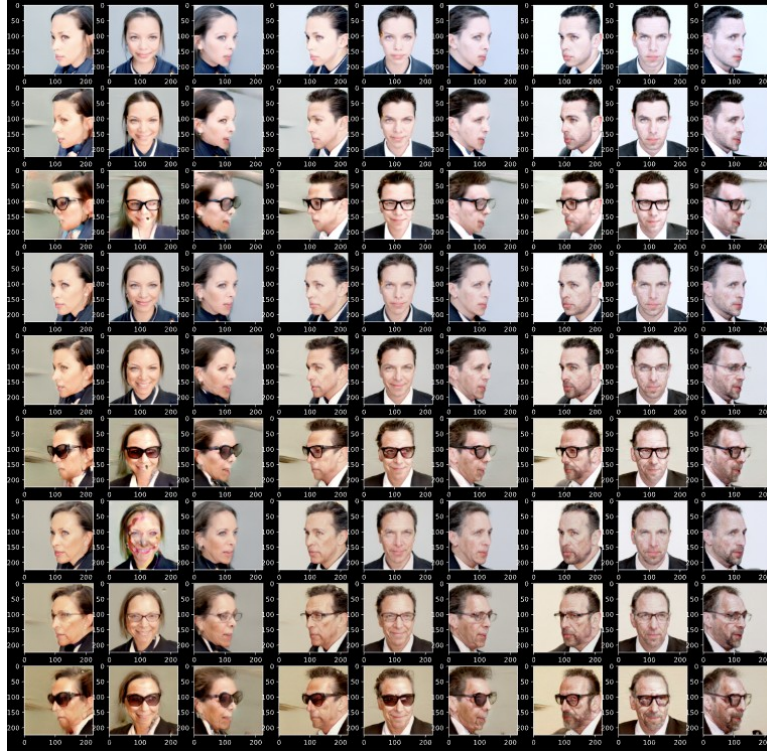


Figure 3: One hybrid-inverted face edited across attribute combinations (eyeglasses, smile, pose): the edits apply cleanly, without the artefacts plain optimisation produced.

4 Results

4.1 Attribute editing

InterFaceGAN edits five attributes: age, eyeglasses, gender, smile and head pose. Sweeping each weight from strongly negative to strongly positive shows robust edits across the useful range and artefacts only at the extremes, as expected. The strips below sample seven points of each sweep; the full animations are in the README.





4.2 Edit, then stylise

We sampled portraits, produced every combination of edited attributes with weights ± 2 , and stylised each with BlendGAN. Reference styles came from the AAHQ test images shipped with BlendGAN and from comics (Hellboy, Watchmen). BlendGAN handled almost every AAHQ style; a few comic references with busy backgrounds and complex textures did not blend cleanly.

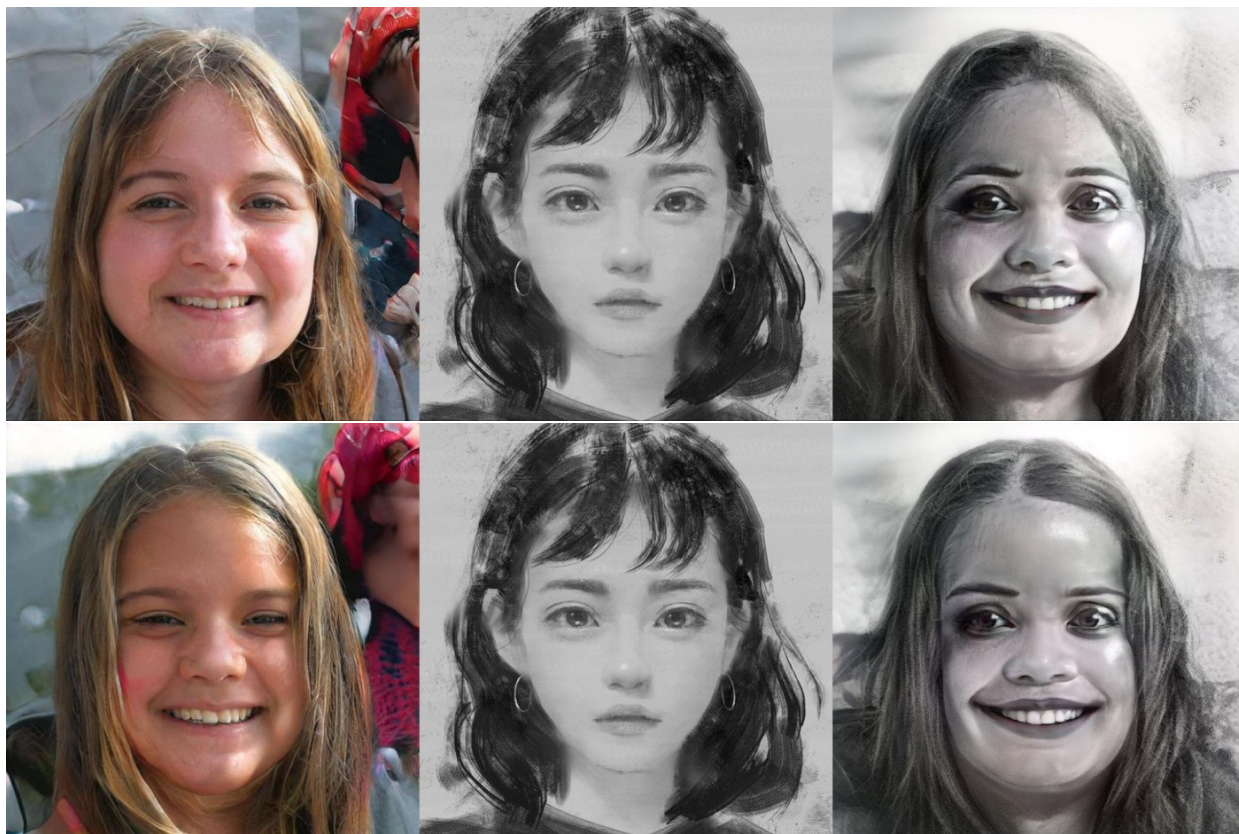




Figure 4: Older, with eyeglasses, across four reference styles (frames of the README animation).

4.3 Pose survives stylisation



5 Conclusion

GANs are effective for image manipulation and stylisation. For real-life use the limiting step is inversion: of the three encoding approaches, the hybrid encoder-plus-optimisation route performed best, at a cost of about five minutes per image and an encoder that is expensive to train. Given an editable code, InterFaceGAN’s edits are robust and BlendGAN preserves them under stylisation across a wide range of reference styles.

6 References

- Y. Shen, J. Gu, X. Tang, B. Zhou. *Interpreting the Latent Space of GANs for Semantic Face Editing*. CVPR 2020. arXiv:1907.10786.
- Y. Shen, C. Yang, X. Tang, B. Zhou. *InterFaceGAN: Interpreting the Disentangled Face Representation Learned by GANs*. TPAMI 2020. arXiv:2005.09635.
- M. Liu et al. *BlendGAN: Implicitly GAN Blending for Arbitrary Stylized Face Generation*. NeurIPS 2021. arXiv:2110.11728.
- W. Xia et al. *GAN Inversion: A Survey*. arXiv:2101.05278.
- E. Richardson et al. *Encoding in Style: a StyleGAN Encoder for Image-to-Image Translation*. CVPR 2021.
- Code used: github.com/genforce/interfacegan · github.com/onion-liu/BlendGAN.

This PDF is typeset from the 2022 course report; the text is the original with light copy-editing, and animated figures are replaced by frame strips.